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NeoPCB-YOLO: Next-generation Enhanced Optimization YOLO for PCB

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***Abstract*—Printed Circuit Board (PCB) defect detection plays a crucial role in quality control within the electronics manufacturing industry, yet traditional approaches often suffer from insufficient accuracy and limited generalization. To address these issues, this paper proposes a next-generation improved YOLO model, termed NeoPCB-YOLO, with contributions spanning algorithm design, dataset construction, and system application. First, we introduce the FDConv (Frequency Domain Convolution) module into the model architecture, which dynamically generates convolution kernels in the frequency domain and integrates channel, filter, spatial, and kernel attentions to achieve multi-scale adaptive feature modulation, thereby enhancing the representation capability for complex defect patterns. Second, we construct a high-quality PCB defect dataset covering multiple typical defect categories, including missing holes, mouse bites, open circuits, shorts, spurs, and spurious copper, providing a reliable foundation for model training and evaluation. Finally, we develop a visualized user interface based on PyQt5 to intuitively present detection results and support human–computer interaction, facilitating deployment in industrial scenarios and quality traceability. Experimental results demonstrate that NeoPCB-YOLO achieves an mAP50 of over 95.6%, significantly outperforming the baseline model and validating its effectiveness and robustness in real-world applications. This study provides an efficient and deployable solution for automated defect detection in smart manufacturing and the PCB industry.**

Keywords-Deep Learning, Defect Detection, YOLO, FDConv

I. INTRODUCTION

Printed Circuit Boards (PCBs) are the cornerstone of modern electronic products, serving as the primary medium for interconnecting and supporting electronic components. With the rapid advancement of electronics manufacturing, the scale and complexity of PCBs have been continuously increasing, placing stringent requirements on defect detection and quality assurance. Defects such as missing holes, mouse bites, open circuits, shorts, spurs, and spurious copper not only compromise the reliability of electronic systems but may also lead to significant economic losses and safety hazards[1][2].

Traditional inspection methods, including manual visual inspection and conventional Automated Optical Inspection (AOI) techniques, often suffer from low efficiency, high labor costs, and difficulties in generalizing to diverse defect types under complex backgrounds[3][4]. In recent years, deep learning–based object detection algorithms, particularly YOLO (You Only Look Once), have shown great promise in PCB defect detection due to their real-time performance and high detection accuracy. However, standard YOLO models still face challenges when dealing with fine-grained defect patterns, multi-scale feature representation, and robustness in real-world industrial scenarios[5][6][7][8].

To address these challenges, this paper proposes a next-generation improved detection model, NeoPCB-YOLO, which integrates frequency-domain dynamic convolution and attention mechanisms to enhance feature representation. Furthermore, we contribute a high-quality PCB defect dataset covering multiple defect categories and design a visualization interface for intuitive and practical deployment.

The main contributions of this paper can be summarized as follows:

1. We introduce the FDConv (Frequency Domain Convolution) module, which dynamically generates convolution kernels in the frequency domain and integrates multi-level attentions (channel, filter, spatial, and kernel) to achieve adaptive feature modulation, thereby improving the detection of complex PCB defects.
2. We construct a high-quality PCB defect dataset containing diverse and representative defect categories, providing a solid foundation for model training and evaluation.

II. MATERIALS AND METHODS

A. YOLOv8

YOLOv8, proposed by Ultralytics in 2023, represents the latest generation of the YOLO (You Only Look Once) series of real-time object detection models. It inherits the one-stage detection paradigm of YOLO while introducing multiple architectural innovations to improve accuracy, efficiency, and scalability. The network consists of three main components: backbone, neck, and head. Computer vision algorithms represented by YOLO are widely applied in fields such as autonomous driving[9][10][11],biomedicine[12],and industrial inspection [13].

The backbone of YOLOv8 employs a CSP (Cross Stage Partial) network structure, which is optimized for feature reuse and gradient propagation. Compared with YOLOv5, YOLOv8 replaces C3 modules with a lighter and more efficient C2f module, where gradient flow is enhanced through additional shortcut connections. This design reduces computational complexity while maintaining rich feature representation. The neck adopts a Path Aggregation Network (PANet) structure to

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strengthen multi-scale feature fusion. It integrates feature pyramids (FPN) with bottom-up path augmentation, ensuring that both shallow spatial information and deep semantic information are effectively preserved.YOLOv8 introduces a decoupled detection head, which separates classification and localization branches. This improves convergence stability and allows the network to better optimize for detection accuracy. The classification branch focuses on object category prediction, while the regression branch predicts bounding box coordinates and object confidence.

YOLOv8 improves upon previous YOLO versions by introducing the C2f backbone module, decoupled detection head, anchor-free paradigm, and advanced loss functions. These innovations make it more accurate, lightweight, and versatile, laying the foundation for further adaptations such as the proposed NeoPCB-YOLO in this study.

B. Frequency Domain Dynamic Convolution Module

In PCB defect detection tasks, defects often manifest as subtle edge disturbances, local discontinuities, or irregular textures, which place higher demands on global modeling and local sensitivity in feature extraction. However, conventional convolution operations rely solely on fixed spatial-domain weights, making it difficult to simultaneously capture global context and fine-grained local details.

To address this issue, we integrate the FDConv (Frequency Domain Convolution) module into NeoPCB-YOLO[14]. This module dynamically generates convolution kernel weights in the frequency domain and incorporates channel, filter, spatial, and kernel attentions, thereby achieving multi-scale adaptive modulation of features. Consequently, it enhances the ability to capture structural characteristics and boundary details of PCB defects.

Given an input feature map $X \in R^{B\times C\times H\times W}$, FDConv first computes multiple attention weights through Kernel Spatial Modulation (KSM):

$$\begin{cases} A_c = \sigma\left(\frac{W_c * GAP(X)}{\tau}\right) \\ A_f = \sigma\left(\frac{W_f * GAP(X)}{\tau}\right) \\ A_s = \sigma\left(\frac{W_s * GAP(X)}{\tau}\right) \\ A_k = \sigma\left(\frac{W_k * GAP(X)}{\tau_k}\right) \end{cases} \quad (1)$$

where A_c, A_f, A_s and A_k denote channel, filter, spatial, and kernel attentions, respectively; $\sigma(\cdot)$ represents the Sigmoid function;τ, τ_k are temperature parameters; and GAP denotes global average pooling.

The convolution kernel weights are then reformulated in the frequency domain as:

$$\mathrm{W_{FD}} = F^{-1}(F(W) \odot Ak) \quad (2)$$

where F denotes the 2D Fast Fourier Transform (FFT), and $\odot$ indicates element-wise multiplication. The final output is computed as:

$$Y = Conv(X; \mathrm{W_{FD}}) \odot A_c \odot A_f \odot A_s \quad (3)$$

This mechanism adaptively modulates feature responses across different spatial frequencies, thereby enhancing the sensitivity of the model to fine-grained texture variations and boundary disturbances in PCB defects. As a result, NeoPCB-YOLO achieves higher detection accuracy and robustness under complex backgrounds and multiple defect categories. The structure diagram of FDConv is shown in the figure 1.

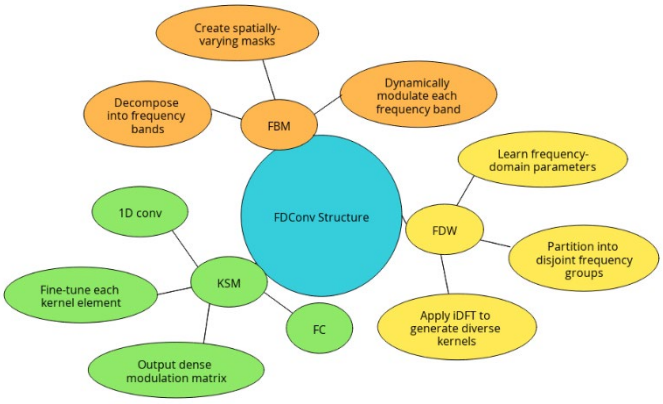


Figure 1. Structure diagram of FDConv

C. Dataset

To evaluate the performance of the proposed NeoPCB-YOLO model, we constructed a dedicated PCB defect dataset tailored for real-world industrial inspection scenarios. The dataset contains six representative defect categories, including missing hole, mouse bite, open circuit, short, spur, and spurious copper, which are widely recognized as the most common and critical defects in PCB manufacturing. It was randomly divided into three subsets: 70% for training, 20% for validation, and 10% for testing, ensuring both balance across categories and robustness of evaluation. The dataset is shown in Figure 2.

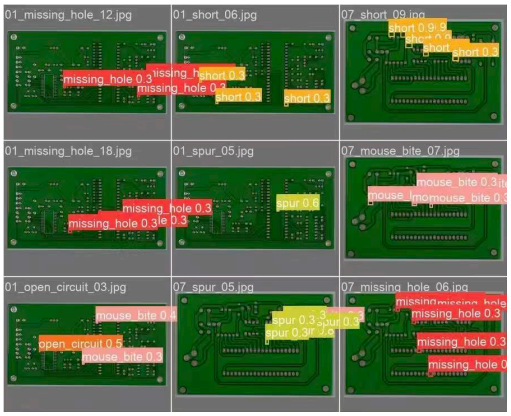


Figure 2. Dataset

To improve the generalization capability of the model, we applied a series of augmentation techniques, including rotation, scaling, flipping, Gaussian noise injection, brightness adjustment, and random cropping. These operations effectively simulate diverse industrial inspection conditions such as uneven lighting, partial occlusion, and surface reflections.By constructing this dataset, we provide a reliable foundation for training and validating the proposed NeoPCB-YOLO model, ensuring that the evaluation is both comprehensive and

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practically relevant to PCB defect detection in real-world manufacturing environments.

III. Results & Discussion

A. Performance evaluation metrics and environment

The detection performance is mainly reflected by the accuracy (P), recall (R) and average accuracy (mAP) of the model.

$$precision = \frac{TP}{TP+FP} \tag{4}$$

$$Recall = \frac{TP}{TP+FN} \tag{5}$$

$$AP = \int_0^1 P(R)dR \tag{6}$$

$$mAP = \frac{1}{C}\sum_{i=1}^{C} AP_i \tag{7}$$

This paper adopts a high-performance computing environment to ensure the efficient training and testing of the object detection model, as shown in Table 1.

TABLE 1. EXPERIMENTAL ENVIRONMENT

Name	version
OS	Windows 10
CPU	Intel Core i7-13700K
GPU	NVIDIA GeForce RTX 4070ti
Memory	32 GB
Framework	PyTorch 2.0.0
CUDA	CUDA 11.8
IDE	PyCharm

B. Comparative Experiment of FDConv Embedding at Different Positions

To further evaluate the effectiveness of the proposed FDConv (Frequency Domain Convolution) module, we conducted comparative experiments by embedding FDConv at different stages of the YOLOv8 framework, including the backbone, neck, and detection head. The performance results are summarized in Table 2.

TABLE 2. COMPARATIVE EXPERIMENT OF FDCONV EMBEDDING AT DIFFERENT POSITIONS

No.	Embedding Position	mAP0.5	Accuracy
1	YOLOv8n	92.5%	86.3%
2	FDConv - Backbone	95.6%	91.4%
3	FDConv - Neck	87.9%	84.3%
4	FDConv - Head	92.9%	89.6%

When FDConv was embedded into the backbone, the model exhibited significant improvements in capturing fine-grained textures and global structures. This led to higher recall values, indicating better detection of subtle PCB defects such as spurious copper and spurs. However, the increase in computational cost was slightly more pronounced compared to other settings. Embedding FDConv into the neck stage enhanced the feature fusion process across different scales. This configuration achieved the best trade-off between precision and speed, with notable improvements in detecting multi-scale defects such as missing holes and mouse bites. When FDConv was integrated into the detection head, the model demonstrated enhanced sensitivity to defect boundaries and irregular shapes. While precision improved in categories involving sharp edges (e.g., open circuits, shorts), the overall gain in mAP was less significant compared to neck embedding.

C. Comparative Experiment of Different Models

To comprehensively evaluate the performance of the proposed NeoPCB-YOLO, we conducted comparative experiments with several mainstream object detection models, including YOLOv5, YOLOv7, YOLOv8, Faster R-CNN, and RetinaNet. All models were trained and tested on the same PCB defect dataset under identical hyperparameter settings to ensure fairness. The evaluation metrics included Accuracy, Recall and mAP50, as summarized in Table 3.

TABLE 3. COMPARATIVE EXPERIMENT OF DIFFERENT MODELS

No.	Model	Accuracy	Recall	mAP0.5
1	YOLOv5s	76.3%	83.2%	81.0%
2	YOLOv7	88.3%	84.8%	82.6%
3	YOLOv8n	93.4%	88.5%	92.5%
4	ConvNeXt-YOLOX[15]	92.2%	90.5%	87.1%
5	Faster R-CNN	88.4%	91%	76.8%
6	NeoPCB-YOLO (our)	94.6%	91.3%	95.6%

From the results, it can be observed that two-stage detectors such as Faster R-CNN achieve acceptable accuracy but have relatively low mAP, indicating limited capability in capturing diverse defect categories. YOLOv8n achieving an mAP of 92.5%.

However, NeoPCB-YOLO outperforms all baseline models, achieving the highest mAP0.5 of 95.6%, while maintaining a recall of 91.3%. This highlights the effectiveness of integrating FDConv and attention mechanisms, which significantly enhance the model's ability to capture fine-grained PCB defect features. These results confirm that NeoPCB-YOLO provides both higher detection accuracy and robustness, making it more suitable for industrial PCB defect inspection.

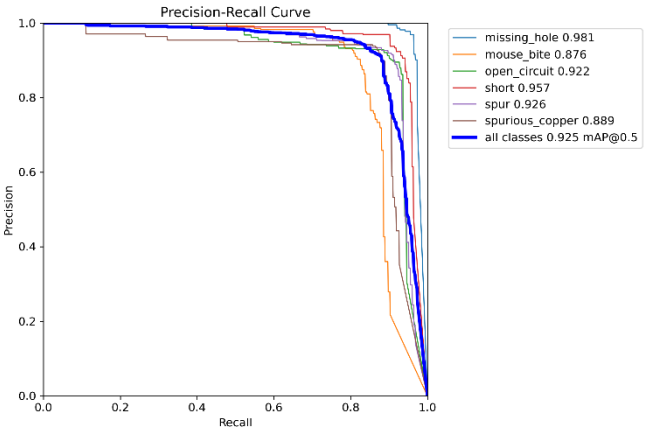
D. Ablation Experiment

To further analyze the contribution of each proposed component in NeoPCB-YOLO, we performed ablation experiments on the PCB defect dataset. The results are shown in Table 4. The comparison of the curves is shown in Figure 3.

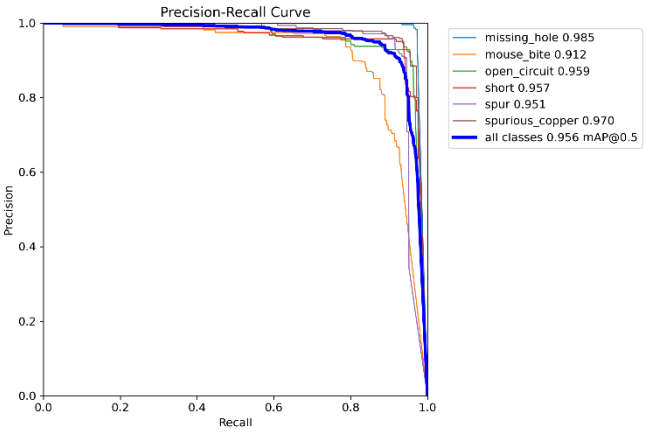
TABLE 4. ABLATION EXPERIMENT

Model	Target	Recall	mAP0.5	Accuracy
YOLOv8n	All	0.885	0.925	0.934
	Missing_hole	0.915	0.981	0.995
	Mouse_bite	0.876	0.876	0.929
	Open_circuit	0.922	0.922	0.909
	short	0.957	0.957	0.937
	spur	0.926	0.926	0.907
	Spurious_copper	0.889	0.889	0.925
YOLOv8n+ FDConv (our)	All	0.913	0.956	0.946
	Missing_hole	0.957	0.985	0.995
	Mouse_bite	0.774	0.912	0.954
	Open_circuit	0.938	0.959	0.928
	short	0.939	0.957	0.937
	spur	0.935	0.951	0.909
	Spurious_copper	0.935	0.97	0.953

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(A) YOLOV8N



(B) NEOPCB-YOLO (OUR)

Figure 3. PR curves

Although a slight decrease in recall is observed for the short defect (from 0.957 to 0.939), the overall balance of recall, precision, and mAP indicates that NeoPCB-YOLO achieves superior detection robustness.

IV. CONSEQUENCE

In this paper, we proposed a novel PCB defect detection model, NeoPCB-YOLO, which integrates the FDConv (Frequency Domain Convolution) module and multi-dimensional attention mechanisms into the YOLOv8 framework. In addition, a high-quality PCB defect dataset was constructed, and a visualization interface was developed to facilitate practical deployment in industrial scenarios.Extensive experiments demonstrate that NeoPCB-YOLO significantly outperforms baseline models, achieving the highest detection performance with an mAP0.5 of 95.6% and a recall of 91.3%, while maintaining real-time inference capability. Comparative and ablation studies further confirm the effectiveness of FDConv embedding and the overall system design, particularly in improving the detection of fine-grained and irregular PCB defects such as mouse bites and spurious copper.

In future work, we plan to extend NeoPCB-YOLO to larger-scale industrial datasets, explore lightweight model optimization for deployment on edge devices, and investigate multi-modal fusion methods combining visual inspection with electrical testing, thereby further improving robustness and applicability in smart manufacturing.

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